

Thesis Part 2: Aims, Objectives, and Methodology

3. AIM AND OBJECTIVES

3.1 Primary Aim

The primary aim of this study is to systematically evaluate the depressive effects of haloperidol when administered at doses sufficient to induce catalepsy in albino mice, thereby establishing a comprehensive understanding of the mood-related consequences of dopaminergic blockade in the context of extrapyramidal symptom development.

3.2 Primary Objectives

1. **To determine the minimum effective dose of haloperidol required to induce significant catalepsy in albino mice** through dose-response studies using standardized catalepsy assessment protocols.
2. **To evaluate the development of depressive-like behaviors following haloperidol administration** using validated behavioral paradigms including the forced swimming test, tail suspension test, and sucrose preference test.
3. **To establish the temporal relationship between catalepsy induction and the onset of depressive-like behaviors** through time-course studies spanning acute and subchronic treatment periods.
4. **To investigate the dose-dependency of haloperidol-induced depressive behaviors** by comparing multiple dose levels and their corresponding effects on mood-related behavioral parameters.

3.3 Secondary Objectives

1. **To assess the reversibility of haloperidol-induced depressive behaviors** following drug discontinuation and determine the recovery timeline for both motor and mood-related effects.
 2. **To investigate potential sex differences in haloperidol-induced catalepsy and depression** by comparing behavioral responses between male and female albino mice.
 3. **To evaluate the correlation between the severity of catalepsy and the magnitude of depressive-like behaviors** to understand the mechanistic relationship between motor and mood effects.
 4. **To assess the impact of haloperidol treatment on general locomotor activity and exploratory behavior** using open field testing to distinguish specific depressive effects from general motor impairment.
 5. **To investigate potential neurochemical correlates of haloperidol-induced depression** through measurement of brain monoamine neurotransmitter levels in relevant brain regions.
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4. PLAN OF WORK

4.1 Study Design Overview

This study employs a comprehensive experimental design incorporating multiple phases to systematically investigate the relationship between haloperidol-induced catalepsy and depression in albino mice. The research plan is structured as a series of interconnected experiments designed to address the primary and secondary objectives outlined above.

4.2 Phase I: Dose-Response Determination

Duration: 4 weeks **Objective:** Establish the dose-response relationship for haloperidol-induced catalepsy

Methodology: Groups of mice will receive varying doses of haloperidol (0.25, 0.5, 1.0, 2.0, and 4.0 mg/kg, i.p.) with catalepsy assessment at multiple time points post-administration.

4.3 Phase II: Depression Behavioral Assessment

Duration: 6 weeks **Objective:** Evaluate depressive-like behaviors at established cataleptic doses

Methodology: Mice will receive haloperidol at doses determined in Phase I, followed by comprehensive behavioral assessment using FST, TST, and sucrose preference tests.

4.4 Phase III: Time-Course Studies

Duration: 8 weeks **Objective:** Determine temporal relationship between catalepsy and depression

Methodology: Longitudinal assessment of both catalepsy and depressive behaviors over acute (single dose) and subchronic (7-day) treatment paradigms.

4.5 Phase IV: Recovery Studies

Duration: 6 weeks **Objective:** Assess reversibility of haloperidol effects **Methodology:** Monitor behavioral recovery following treatment discontinuation over a 21-day washout period.

4.6 Phase V: Neurochemical Analysis

Duration: 4 weeks **Objective:** Investigate neurochemical correlates of behavioral changes **Methodology:** High-performance liquid chromatography (HPLC) analysis of dopamine, serotonin, and norepinephrine levels in relevant brain regions.

5. MATERIALS AND METHODS

5.1 Experimental Animals

Species and Strain: Adult albino mice (*Mus musculus*) of Swiss albino strain **Source:** Central Animal House, [Institution Name] **Age:** 8-12 weeks at the commencement of experiments **Weight Range:** 25-35 grams for males, 20-30 grams for females **Sex Distribution:** Equal numbers of male and female mice in all experimental groups **Housing Conditions:** Animals housed in polypropylene cages (43 × 27 × 15 cm) with stainless steel wire mesh tops, maximum 6 animals per cage **Environmental Conditions:**

Temperature maintained at $22 \pm 2^{\circ}\text{C}$, relative humidity 50-60%, 12:12 hour light-dark cycle (lights on at 6:00 AM) **Acclimatization:** Minimum 7 days acclimatization period before experimental procedures

5.2 Animal Welfare and Ethical Considerations

All experimental procedures were conducted in strict accordance with the guidelines established by the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), Government of India. The experimental protocol received prior approval from the Institutional Animal Ethics Committee (IAEC) under protocol number [Protocol Number]. Animal welfare was monitored throughout the study period, with particular attention to signs of distress, weight loss, or abnormal behavior. Humane endpoints were established to minimize animal suffering, and veterinary consultation was available for any health concerns.

5.3 Drugs and Chemicals

Haloperidol: Pharmaceutical grade haloperidol obtained from [Supplier Name], purity >99% as determined by HPLC analysis. Stock solutions prepared in dimethyl sulfoxide (DMSO) and further diluted in 0.9% saline to achieve final DMSO concentration <1%.

Vehicle Control: 0.9% saline containing <1% DMSO, administered at equivalent volumes to drug-treated groups.

Other Chemicals:

- Sucrose (analytical grade) - [Supplier Name]
- Standard neurotransmitters for HPLC analysis (dopamine, serotonin, norepinephrine) - [Supplier Name]
- Mobile phase constituents for HPLC - [Supplier Name]
- All chemicals were of analytical or HPLC grade quality

5.4 Experimental Groups and Treatment Protocols

Based on the multi-phase design, animals were randomly assigned to experimental groups using a computer-generated randomization sequence. Group assignments were concealed until the completion of behavioral assessments to ensure blinded evaluation.

Phase I Groups (n=8 per group):

- Vehicle control (saline + <1% DMSO)
- Haloperidol 0.25 mg/kg
- Haloperidol 0.5 mg/kg
- Haloperidol 1.0 mg/kg
- Haloperidol 2.0 mg/kg
- Haloperidol 4.0 mg/kg

Phase II Groups (n=12 per group):

- Vehicle control
- Haloperidol at ED50 dose for catalepsy
- Haloperidol at ED75 dose for catalepsy
- Positive control (reserpine 2 mg/kg) - for depression validation

5.5 Drug Administration

All drug administrations were performed via the intraperitoneal (i.p.) route using sterile 26-gauge needles and 1 mL syringes. Injection volumes were standardized at 10 mL/kg body weight to minimize volume-related effects. Drug solutions were prepared fresh on the day of administration and protected from light. Administration time was standardized between 9:00-10:00 AM to minimize circadian rhythm effects on behavioral outcomes.

5.6 Catalepsy Assessment

Catalepsy was assessed using the horizontal bar test, a well-validated method for quantifying drug-induced cataleptic behavior in rodents. The apparatus consisted of a horizontal metal bar (diameter 0.9 cm) positioned 9 cm above a padded surface.

Procedure:

1. Mice were gently lifted and their forepaws placed on the horizontal bar
2. The hind paws remained on the testing surface
3. The duration for which the imposed posture was maintained was recorded
4. A maximum cut-off time of 300 seconds was employed to prevent excessive stress
5. Catalepsy was assessed at 30, 60, 120, and 240 minutes post-drug administration
6. Animals were considered cataleptic if they maintained the imposed posture for > 30 seconds

Scoring Criteria:

- 0: Immediate correction of posture (<5 seconds)
- 1: Mild catalepsy (5-30 seconds)
- 2: Moderate catalepsy (30-120 seconds)
- 3: Severe catalepsy (> 120 seconds)

5.7 Depression Behavioral Assessments

5.7.1 Forced Swimming Test (FST)

The FST was conducted using transparent cylindrical tanks (height: 40 cm, diameter: 18 cm) filled with water to a depth of 30 cm at $25 \pm 1^\circ\text{C}$. This water depth prevented animals from touching the bottom

with their hind paws or tail, ensuring genuine swimming behavior.

Procedure:

1. Mice were individually placed in the water-filled cylinder
2. Behavior was recorded for 6 minutes using video cameras positioned to provide clear side views
3. The first 2 minutes served as an adaptation period
4. Immobility time during the final 4 minutes was analyzed
5. Immobility was defined as the absence of all movement except minimal motions required to keep the head above water
6. Sessions were conducted 24 hours after drug administration in the main study
7. Water was changed between each animal to prevent pheromonal effects

Behavioral Parameters:

- Total immobility time (seconds)
- Latency to first immobility episode (seconds)
- Number of immobility episodes
- Swimming behavior intensity (scored 0-3)
- Climbing behavior intensity (scored 0-3)

5.7.2 Tail Suspension Test (TST)

The TST apparatus consisted of a horizontal bar positioned 50 cm above the ground with provisions for suspending mice by their tails using adhesive tape.

Procedure:

1. Mice were suspended by their tails using adhesive tape placed 2 cm from the tip
2. The suspension height was adjusted so that the mouse's head was 20 cm from the ground
3. Behavior was recorded for 6 minutes
4. Immobility time during the entire 6-minute period was analyzed
5. Immobility was defined as the absence of limb movement and body swaying
6. Testing was conducted in a quiet, dimly lit environment
7. Sessions were spaced appropriately to avoid stress accumulation

Behavioral Parameters:

- Total immobility time (seconds)
- Latency to immobility (seconds)
- Duration of active struggling (seconds)

- Number of immobility bouts

5.7.3 Sucrose Preference Test

This test assessed anhedonic behavior through measurement of the animal's preference for a palatable sucrose solution over plain water.

Procedure:

1. Training phase: Mice were habituated to 1%